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PI Name:	Orasanu, Judith M. Ph.D.		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
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Space Biology Special Category:	None		
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Contact Monitor:		**Contact Phone:**	
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	AIMS: Since the beginning of human space flight, psychological, behavioral, and interpersonal aspects of crew performance have jeopardized crew health, safety and mission success. The goal of this project is to optimize team psychosocial adaptation and performance in exploration missions (Risk #24, NASA Critical Path Roadmap). We aim to enhance the ability of crews to respond adaptively to unanticipated problems, to maintain positive social and task cohesion, and to collaborate effectively under stress. The research will (a) determine the effects of task and interpersonal stressors on team performance and cohesion; (b) identify effective team interaction and stress management strategies; (c) determine the effects of individual traits and team composition on team cohesion and performance; and (e) develop countermeasures to prevent deterioration of team performance and mitigate the effects of stress. APPROACH: A teamwork simulation laboratory has been created at NASA Ames Research Center to address these issues. Teams composed of 4 or 5 members engage in Antarctic or Lunar search missions over a period of 4 days (one		

Task Description:	training day and 6 simulations over 3 days). Social and task stressors are manipulated along with team composition. Time-synchronized data (computer-based task performance, physiological stress measures, audio and video recordings) support analyses of team processes and outcomes. Questionnaire data that assess workload, team dynamics, personality, and other individual characteristics provide essential information concerning team composition, leadership and team interaction. Cognitive functioning tests assess transient variations in mental fitness. KEY FINDINGS: During Year 3 data collection was completed for a study of training effects on team cohesion and performance in gender-mixed crews. Further analyses were performed on data collected in our prior study involving teams of 4 males. Analyses of both data sets address (a) behavioral and social stressors that affect or contribute to the breakdown of individual and team performance and mission support coordination, and (b) factors that enhance the crew's ability to live and work in the space environment. Two operational and social stressors were found to affect individual and team performance: task stress and negative social dynamics. Task stress reflected mission difficulty, manipulated through time pressure, workload, ambiguity of diagnostic cues, and impediments to mission success. Performance was significantly worse on the three difficult missions than on the three moderate ones. Performance improved significantly over the three mission days, but was steady only for the moderate scenarios, with continued improvement from day 2 to day 3, while performance on the difficult missions dropped on day 3. The second stressor relating to performance involved negative interpersonal dynamics. High levels of Anger and Aggression as measured by the Group Environment Scale were negatively correlated with team success reflected in mean team "points" awarded for successfully processing elemental tasks within each mission. High levels of Innovation, which measures the extent to which the crew "tries out new things," also were negatively related to team performance. We suspect this reflected the teams' floundering in search of successful teamwork strategies. Conversely, high Order and Organization scores were positively associated with team performance. Order and Organization scores were negatively correlated with Anger and Aggression scores, which were positively correlated with Innovation scores. Positive factors that contributed to individual and team mission success include team training, team self-management or macrocognitive strategies, and observed social interaction style. Interpersonal Relations Training (IRT) was designed to foster crewmember trust, cohesion, and self-awareness, factors that have been found in high performing teams. This approach was contrasted with Team Adaptation and Coordination Training (TACT) designed to train crews to work together as a team, cope with stress, and communicate effectively. TACT training led to marginally better individual and team performance than IRT training. Crewmembers who received TACT training earned almost 50% more points than IRT trained individuals; they also completed twice as many missions as IRT-trained crews. One possible reason the IRT training did not enhance performance was that it did not achieve its goal of improved team cohesion, which did not differ significantly across training conditions. One of our prior studies found that positive team cohesion was associated with high levels of team coordination and mission success. More successful teams also exhibited high levels of "macrocognitive" strategies. These are team self-management strategies including resource management, reliance on support capabilities of the base coordinator, multitasking, and contingency planning. The other feature associated with successful team performance was positive affect expressed in crew communication. This includes joking and compliments on each other's performance, and is measured as the ratio of positive affect to total communication. IMPLICATIONS: These findings indicate that negative team dynamics are incompatible with effective individual and team performance. Techniques to enhance positive team cohesion and reduce negative affect will be essential for both crew psychosocial health and high levels of performance. Team adaptation and coordination training can overcome the stress associated with task difficulty through improved team self-management strategies. Results will serve as the basis for selecting, composing and training teams to assure long-term health and successful performance. PLANS: Preliminary work was done for our next study in which we will examine the effects of fatigue on crew interaction patterns and performance using our team monitoring techniques.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Most work in government and industry is conducted in teams at least a significant portion of the time. Better understanding of factors that contribute to effective and cohesive team performance for space crews will have numerous benefits, especially in high-risk industries, government, and the military. These benefits include greater efficiency, fewer errors, greater productivity, and less time lost due to interpersonal frictions or stress in the workplace. Our study will indicate which instruments are valid and useful measures of team performance and group cohesion. Understanding how to measure performance, productivity, and cohesion is the first step to being able to predict those factors. Findings on optimal crew composition will be beneficial in the design of industry work teams or any other teams in which members must work closely in a collaborative environment. In addition, our study will yield information on conflict management strategies and validated team training that could be built into education programs in colleges, professional schools, medical training, and preparation for crisis-management teams (e.g., negotiators, firefighters, police, or disaster relief teams). Countermeasures that result from the present project will have broad application to teams that operate under stressful conditions such as flight crews, air traffic controllers, medical personnel, offshore oil workers, or nuclear power operators.
	Two major tasks were accomplished during Year Three: data collection for Study 3 and further analyses of data from Studies 2 and 3. Study 3: Data collection was completed for a study of the effects of training on crews' ability to cope with interpersonal and task stressors and to work effectively as a team. This experiment involved 5-member gender-mixed crews of 2 females and 3 males each, randomly assigned to one of four experimental conditions (N = 120). Two subteams of 2 people searched for water on the lunar surface, supported by a base coordinator who received updated satellite information which had to be evaluated and shared with the search teams, and who assisted the crews to refuel and resupply.

Task Progress:	Two training and two control conditions were compared: (1) Interpersonal Relations Training (IRT) involving 22 hours of team exercises, didactic instruction, and feedback designed to build trust among crewmembers, self-understanding, and team cohesion; (2) Team Adaptation and Coordination Training (TACT), an 8-hour program emphasizing information sharing, planning, team monitoring and feedback; (3) a team familiarity control (FC) condition involving 22 hours of live action role play but no training, and (4) a no-training/no-familiarity control condition. Task difficulty was varied within crews (3 difficult and 3 moderate missions over 3 days, one difficult/one moderate per day). Individual and team performance measures were collected as well as physiological arousal, audio/video recordings, personality, cognitive functioning and team dynamics measures. Data from Study 3 were analyzed to assess the following: 1. Individual and crew-level mission performance as a function of training condition, task difficulty, day of mission, and time of day of study 2. Training effects on team cohesion and other aspects of crew dynamics as measured by the Group Environment Scale 3. The relation between crew dynamics and individual and team mission performance 4. Predictive validities were determined for the Spaceflight Cognitive Assessment Tool and the Psychomotor Vigilance Test, which were administered prior to each mission. Analyses determined whether their predictive validities were constant across days, mission difficulty and time of day, as well as the relative efficacy of various subscales. Data from prior studies involving all-male, all-female, or gender mixed crews were further analyzed: 1. Pre-simulation strategy planning was analyzed in relation to task performance and adaptation across the 6 missions. 2. Task and social aspects of crews' communication were analyzed in relation to team success. We developed a procedure to code relational aspects of communication directly from audiotapes without first transcribing the tapes, an arduous and costly task. 3. Psychophysiological data were cleaned in preparation for analyzing their relation to task difficulty, subjective workload, transient task stressors and personality.
Bibliography Type:	Description: (Last Updated: 05/21/2010)
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